

MATH 231 – Linear Algebra I

Homework 2 ■ Norms, Bases, $\mathbb{R}^n$, and the Gradient

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Due: to be announced on the class Discord.

Lecture	Sections	Topics
Vector Unit, Part 2	§10–§12	<i>revisited:</i> normalization and unit vectors; the direction–magnitude decomposition; projection and the orthogonal piece, used in new ways
Vector Unit, Part 3	§13–§15	Cauchy–Schwarz and its equality case; the metric axioms; the Manhattan and Chebyshev metrics; the p -norm and $p \rightarrow \infty$; unit balls; the norm axioms N1–N3
Vector Unit, Part 4	§16–§20	the standard basis; linear combinations, span, basis, coordinates; dimension; linear independence; subspaces; $\mathbb{R}^3$
Vector Unit, Part 5	§21–§24	orthogonal vectors are independent; coordinates by projection; making a basis orthogonal; the Pythagorean identity; $\mathbb{R}^n$ and every definition in Σ notation
Vector Calculus Unit, Part 1	§1–§7	the derivative and its rules; e^x and $\ln x$ and their derivatives; functions of two variables; partial derivatives; the gradient

Grading. Every problem is worth the same number of points.

On the proof problems (A.1(d), A.4(c)–(d), A.5, A.6(a)–(c), B.3, B.4, B.5, the justifications in C.4, C.5, C.6, D.3, D.4, D.5, E.2, E.3(c), E.5(a), E.7, F.3, and F.6(c)–(d)). None of them requires a clever idea. What they require is that you write down what you are assuming, work in components or from the stated laws, justify each equality by naming the fact that licenses it, and end with a sentence saying what you have shown. A chain of equalities with no words is not a proof; neither is a paragraph of words with no computation. Several of them ask you *not* to use components: that restriction is the point of the problem, because an argument that uses only the laws works in $\mathbb{R}^n$ unchanged.

Collaboration is encouraged — work at a board together, argue about the problems. What you submit must be written by you, in your own words.

What to submit

Push your written work for Parts A–F to your own MATH 231 GitHub repository, as a single PDF named `hw2.pdf` in a folder named `homework/hw2/`. Handwritten and scanned or photographed is completely fine, as long as it is legible and the pages are in order.

PDF is the only accepted format. A `.docx`, a `.jpg`, a `.png`, a `.heic`, or a link to a file hosted somewhere else will not be graded. Phone scanning apps export PDF directly, and every word processor can “Save as PDF”.

Type Parts A–F up instead and you earn +5 points of extra credit on this assignment. Any typesetting system counts — L^AT_EX, Word’s equation editor, or anything else that produces readable mathematics — and the extra credit is for the whole of Parts A–F, not for typing up some of it. The file you submit must still be a PDF.

If you have not set up the repository yet, Setup Guide 1 (§ “Create your folders” and § “Save your work and send it to GitHub”) is the walkthrough. Only pushed work can be graded.

Label every answer

Label each answer with the number of the problem it answers — A.3(d), C.2(e), F.4(h) — and put your answers **in the same order as they appear in this assignment**. A grader who cannot tell which question a piece of work belongs to cannot award it credit, and that is the most common way a correct answer loses points.

You do not have to copy the question out. This applies to handwritten and typed submissions alike: the label alone is enough. Write A.3(d), then your work. Do not reproduce the wording of the problem.

Part A. Dot product and projection, revisited

Vector Unit, Part 2. For $\mathbf{u} \neq \mathbf{0}$,

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}, \quad \text{comp}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|}, \quad \text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \mathbf{u},$$

and the laws of the dot product are **D1** symmetry, **D2** additivity, **D3** homogeneity, **D4** positive definiteness, introduced in §11.8 (“The algebraic laws of the dot product”) of the Vector Unit, Part 2 lecture, *Orientation, the Dot Product, and Projection*.

Problem A.1 (Unit vectors and the direction–magnitude decomposition).

- (a) Normalize each vector below and check that the result has norm 1. Then write each vector in the direction–magnitude form $\mathbf{u} = \|\mathbf{u}\| \hat{\mathbf{u}}$ of §12.2:

$$[-5, 12], \quad [2, -2], \quad [0, -7], \quad [\sqrt{3}, 1].$$

- (b) Write down the vector of length 10 pointing the same way as $[3, -4]$, and the vector of length 3 pointing *opposite* to $[1, 2]$. Build each one as (scalar) $\times$ (unit vector).

(c) Recall from §12.1 that $\mathbb{S}^1$ is the **unit circle**, the set of all unit vectors in the plane:

$$\mathbb{S}^1 = \{ \mathbf{u} \in \mathbb{R}^2 : \|\mathbf{u}\| = 1 \}.$$

Which of $[0.6, -0.8]$, $[\frac{1}{2}, \frac{1}{2}]$, $[-1, 0]$ and $[\cos 2, \sin 2]$ lie in $\mathbb{S}^1$? Justify each.

(d) Let $\mathbf{u} \neq \mathbf{0}$ and $\alpha \neq 0$. Show that normalizing $\alpha\mathbf{u}$ gives $\hat{\mathbf{u}}$ when $\alpha > 0$ and $-\hat{\mathbf{u}}$ when $\alpha < 0$. Which rule from §8 does the work?

Problem A.2 (Solving for an unknown component).

Find every real number k satisfying each condition. Show the equation you solve.

(a) $[k, 3]$ is orthogonal to $[2, -4]$.

(b) $[k, 1]$ is orthogonal to $[k, -4]$.

(c) The angle between $[k, 1]$ and $[1, 0]$ is $\pi/3$.

(d) $\cos([1, k], [3, 3]) = \frac{1}{2}$.

Watch out. In (c) and (d) you will want to square both sides. Squaring can create a “solution” of the squared equation that fails the original one. Check every candidate in the original equation and say why you discard any that fail.

Problem A.3 (Work, and a box on a ramp).

In physics a force is a *vector* $\mathbf{F} \in \mathbb{R}^2$: its direction is the way it pushes, and its norm $\|\mathbf{F}\|$ is how hard. Call a force *constant* when it is the same vector at every moment of the motion. If a constant force acts on an object that moves along a straight line, from its starting point to its finishing point, the *displacement* is the vector $\mathbf{s} \in \mathbb{R}^2$ from start to finish, and the **work** done by the force is the scalar

$$W = \langle \mathbf{F}, \mathbf{s} \rangle,$$

a single real number — in joules, when $\mathbf{F}$ is in newtons and $\mathbf{s}$ in metres. So in this problem $\mathbf{F}$, $\mathbf{s}$, $\mathbf{d}$, $\mathbf{G}$ and $\mathbf{P}$ are vectors in $\mathbb{R}^2$, while W , every norm, and every component such as $\text{comp}_{\mathbf{d}}(\mathbf{G})$ are scalars. A box sits on a ramp that rises in the direction $\mathbf{d} = [4, 3]$, and gravity pulls on it with the constant force $\mathbf{G} = [0, -50]$.

(a) Compute $\text{comp}_{\mathbf{d}}(\mathbf{G})$, $\text{proj}_{\mathbf{d}}(\mathbf{G})$, and the orthogonal piece $\mathbf{G} - \text{proj}_{\mathbf{d}}(\mathbf{G})$. Check that the orthogonal piece is orthogonal to $\mathbf{d}$.

(b) Compute the norms of the two pieces. One of them pulls the box down the ramp and the other presses it into the ramp surface. Which is which?

(c) The box is pushed from $(0, 0)$ to $(8, 6)$. Compute the work done by gravity, and explain what its sign means.

(d) Over the same displacement a worker pushes with $\mathbf{P} = [30, 10]$. Compute the work two ways: as $\langle \mathbf{P}, \mathbf{s} \rangle$, and as $\text{comp}_{\mathbf{s}}(\mathbf{P}) \cdot \|\mathbf{s}\|$. Explain why the second formula says that only the part of $\mathbf{P}$ along $\mathbf{s}$ does work.

- (e) Of all forces of magnitude 40, which one does the most work over this displacement, and how much? Name a nonzero force that does no work at all. Justify both with $\langle \mathbf{F}, \mathbf{s} \rangle = \|\mathbf{F}\| \|\mathbf{s}\| \cos \theta$.

Problem A.4 (No cancellation, and what an equation of dot products says).

Let $\mathbf{u} = [2, -1]$.

- (a) Find three different vectors $\mathbf{v}$ with $\langle \mathbf{u}, \mathbf{v} \rangle = 5$.
- (b) Compute $\text{proj}_{\mathbf{u}}(\mathbf{v})$ for each of your three. What do you notice? Describe the set of *all* $\mathbf{v}$ with $\langle \mathbf{u}, \mathbf{v} \rangle = 5$ geometrically.
- (c) Prove that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$: if $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle$, then $\mathbf{u}$ is orthogonal to $\mathbf{v} - \mathbf{w}$. Use only **D1–D4**.
- (d) Prove: if $\langle \mathbf{x}, \mathbf{v} \rangle = \langle \mathbf{x}, \mathbf{w} \rangle$ for *every* $\mathbf{x} \in \mathbb{R}^2$, then $\mathbf{v} = \mathbf{w}$.

Hint. You are allowed to choose $\mathbf{x}$. One choice finishes the proof in two lines using **D4**.

Problem A.5 (The parallelogram law, polarization, and two theorems from Euclid).

In (a)–(d), work from **D1–D4** and $\langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$ only — no components — so that each argument holds in $\mathbb{R}^n$ as written.

- (a) **Parallelogram law.** Prove $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.
- (b) **Polarization.** Prove $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$.
- (c) Prove $\langle \mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$, and deduce that *the diagonals of a rhombus are orthogonal*. (A rhombus is a parallelogram with four equal sides, which is the case $\|\mathbf{u}\| = \|\mathbf{v}\|$.)
- (d) **Thales' theorem.** Let $\mathbf{a} \neq \mathbf{0}$ and let $\mathbf{p}$ be any vector with $\|\mathbf{p}\| = \|\mathbf{a}\|$ and $\mathbf{p} \neq \pm \mathbf{a}$. So $-\mathbf{a}$, $\mathbf{a}$ and $\mathbf{p}$ are three points on the circle of radius $\|\mathbf{a}\|$ centred at the origin, and $-\mathbf{a}$, $\mathbf{a}$ are the ends of a diameter. Use (c) to prove that $\mathbf{p} - \mathbf{a}$ is orthogonal to $\mathbf{p} + \mathbf{a}$. Then say which angle of which triangle has been shown to be a right angle.
- (e) Check (a) and (b) numerically on $\mathbf{u} = [3, 1]$ and $\mathbf{v} = [-1, 2]$, and check (d) on $\mathbf{a} = [5, 0]$, $\mathbf{p} = [3, 4]$.

Problem A.6 (Three facts about projection).

Let $\mathbf{u} \neq \mathbf{0}$. Prove each statement for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and scalars α, β , using the formula for $\text{proj}_{\mathbf{u}}(\mathbf{v})$ and **D1–D4**.

- (a) **Only the line matters.** For every $\alpha \neq 0$, $\text{proj}_{\alpha \mathbf{u}}(\mathbf{v}) = \text{proj}_{\mathbf{u}}(\mathbf{v})$. But $\text{comp}_{\alpha \mathbf{u}}(\mathbf{v})$ equals $\text{comp}_{\mathbf{u}}(\mathbf{v})$ when $\alpha > 0$ and $-\text{comp}_{\mathbf{u}}(\mathbf{v})$ when $\alpha < 0$.
- (b) **A shadow of a shadow.** $\text{proj}_{\mathbf{u}}(\text{proj}_{\mathbf{u}}(\mathbf{v})) = \text{proj}_{\mathbf{u}}(\mathbf{v})$. Say what this means geometrically.
- (c) **Projection is linear in what is projected.** $\text{proj}_{\mathbf{u}}(\mathbf{v} + \mathbf{w}) = \text{proj}_{\mathbf{u}}(\mathbf{v}) + \text{proj}_{\mathbf{u}}(\mathbf{w})$ and $\text{proj}_{\mathbf{u}}(\beta \mathbf{v}) = \beta \text{proj}_{\mathbf{u}}(\mathbf{v})$.
- (d) **But not in the direction projected onto.** Show by computing that $\text{proj}_{\mathbf{u} + \mathbf{w}}(\mathbf{v}) \neq \text{proj}_{\mathbf{u}}(\mathbf{v}) + \text{proj}_{\mathbf{w}}(\mathbf{v})$ for $\mathbf{u} = [1, 0]$, $\mathbf{w} = [1, 1]$, $\mathbf{v} = [0, 1]$.

Part B. Cauchy–Schwarz, metrics, and the family of norms

Vector Unit, Part 3. For $\mathbf{v} = [v_1, v_2]$ and $p \geq 1$,

$$\|\mathbf{v}\|_1 = |v_1| + |v_2|, \quad \|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2}, \quad \|\mathbf{v}\|_\infty = \max\{|v_1|, |v_2|\}, \quad \|\mathbf{v}\|_p = (|v_1|^p + |v_2|^p)^{1/p},$$

and each norm gives a metric $d_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p$. A bare $\|\mathbf{v}\|$ still means $\|\mathbf{v}\|_2$. The metric axioms are **M1–M4** (§14.1), the norm axioms **N1–N3** (§15.1). Leave answers exact unless a decimal is asked for.

Problem B.1 (The family of norms, by hand).

Let $\mathbf{v}$ range over the four vectors

$$(a) [-3, 4], \quad (b) [5, -12], \quad (c) [2, 2], \quad (d) [-7, 0],$$

- (i) Compute $\|\mathbf{v}\|_1$, $\|\mathbf{v}\|_2$ and $\|\mathbf{v}\|_\infty$, and confirm the ordering $\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1$ of §14.8.
- (ii) For one of the four, all three norms are equal. Which one, and what feature of the vector forces it?
- (iii) For vector (a), compute $\|\mathbf{v}\|_p$ for $p = 3, 4$ and 10 , to four decimal places. The values should be decreasing toward $\|\mathbf{v}\|_\infty$. Compute the upper bound $2^{1/p}M$ from the squeeze argument of §14.7 at $p = 10$, where $M = \|\mathbf{v}\|_\infty$, and confirm that $\|\mathbf{v}\|_{10}$ lies between M and that bound.

Problem B.2 (Which customer is nearest? It depends on the metric).

A depot sits at $P = (1, -1)$, and three customers at $A = (7, -1)$, $B = (6, 1)$ and $C = (5, 3)$.

- (a) Make a table of d_1 , d_2 and d_∞ from P to each customer.
- (b) Three vehicles leave the depot: a *taxi* that must follow a rectangular street grid, a *drone* that flies in a straight line, and an *overhead crane* whose east–west and north–south motors run at the same time and the same speed. Which metric measures the trip for each vehicle? Which customer is nearest for each?
- (c) On one set of axes, sketch the three “circles” of radius 6 centred at P — the sets $\{\mathbf{x} : d(P, \mathbf{x}) = 6\}$ for $d = d_1, d_2, d_\infty$ — and plot A, B, C .

Problem B.3 (The Manhattan and Chebyshev metrics are metrics).

A function d that takes two vectors of $\mathbb{R}^2$ and returns a real number is a **metric** on $\mathbb{R}^2$ when it satisfies the metric axioms of §14.1. The three you need here are, for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^2$:

M1 $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$ zero distance only from a point to itself

M2 $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ symmetry

M3 $d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \geq d(\mathbf{x}, \mathbf{z})$ the triangle inequality

§14.3 left the first of the two metrics below as an exercise. Prove both directly from the definitions and the scalar triangle inequality $|a + b| \leq |a| + |b|$ for real numbers — not by citing §15.2.

- (a) Prove that d_1 satisfies **M1**, **M2** and **M3** on $\mathbb{R}^2$.

(b) Prove the same for d_∞ .

Hint. For **M3**, fix a coordinate i and show $|x_i - z_i| \leq d_\infty(\mathbf{x}, \mathbf{y}) + d_\infty(\mathbf{y}, \mathbf{z})$. Since this holds for $i = 1$ and for $i = 2$, it holds for the larger of the two.

Problem B.4 (When Cauchy–Schwarz is an equality).

§13 showed, using the angle θ , that equality in $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ happens exactly for parallel or antiparallel vectors, and the remark “Cauchy–Schwarz, one more time” in §23.4 settled it again using the Pythagorean identity. Prove it a third way, from **D1–D4** alone — no angle, and no appeal to Pythagoras — so that nothing but the laws is needed.

(a) Show that if $\mathbf{v} = \alpha \mathbf{u}$ for some scalar α , then $|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\|$.

(b) Now suppose $\mathbf{u} \neq \mathbf{0}$ and $|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\|$. Put $c = \langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{u}\|^2$. Expand $\|\mathbf{v} - c\mathbf{u}\|^2$ using **D1–D4** and show that

$$\|\mathbf{v} - c\mathbf{u}\|^2 = \|\mathbf{v}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{u}\|^2}.$$

Conclude that $\mathbf{v} = c\mathbf{u}$.

Problem B.5 (The reverse triangle inequality).

Let $\|\cdot\|$ be *any* norm on $\mathbb{R}^2$ — you may use only **N1–N3**. Prove that for all $\mathbf{u}, \mathbf{v}$,

$$|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u} - \mathbf{v}\|.$$

Hint. Write $\mathbf{u} = (\mathbf{u} - \mathbf{v}) + \mathbf{v}$ and apply **N3** to get one inequality. Swap the roles of $\mathbf{u}$ and $\mathbf{v}$ for the other, and use **N2** with $\alpha = -1$ to see that $\|\mathbf{v} - \mathbf{u}\| = \|\mathbf{u} - \mathbf{v}\|$.

Part C. Bases, span, independence, and subspaces

Vector Unit, Part 4. Recall that $\alpha \mathbf{u} + \beta \mathbf{v}$ is a **linear combination**, the **span** is the set of all of them, and a **basis** spans without redundancy (§16.3). Two vectors are **linearly independent** when $\alpha \mathbf{u} + \beta \mathbf{v} = \mathbf{0}$ forces $\alpha = \beta = 0$ (§18.4); for more vectors the definition is the same with more terms (the table in §24.2). For a pair in $\mathbb{R}^2$, the number $D = u_1 v_2 - u_2 v_1$ decides everything: when $D \neq 0$ the coordinates of $\mathbf{w}$ are

$$\alpha = \frac{w_1 v_2 - w_2 v_1}{D}, \quad \beta = \frac{u_1 w_2 - u_2 w_1}{D} \quad (\S 18.5).$$

§18.5 proves that an independent pair has $D \neq 0$. The converse holds too, and you may use it: if $D \neq 0$, apply the elimination of §18.5 with $\mathbf{w} = \mathbf{0}$, which gives $\alpha D = 0$ and $\beta D = 0$, so the only combination producing $\mathbf{0}$ is the trivial one. A **subspace** is a subset that is non-empty, closed under addition, and closed under scalar multiplication (§19.3).

Watch out. The lecture labels the three subspace conditions S1–S3. Those are *not* the scalar-multiplication laws S1–S4 of §8.4. To avoid the clash, this assignment refers to them by name: non-empty, closed under addition, closed under scaling.

Problem C.1 (Coordinates against a basis that is not the standard one).

Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$.

- (a) Compute D and explain why $\{\mathbf{u}, \mathbf{v}\}$ is a basis of $\mathbb{R}^2$.
- (b) Find the coordinates of $\mathbf{w} = [5, 5]$ and of $\mathbf{w} = [1, -2]$ by setting up the linear system and eliminating by hand, as in §18.1.

Problem C.2 (Independent or dependent?).

For each pair, decide whether it is linearly independent. Argue from the definition — set up $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$ and solve — and confirm with D . For each *dependent* pair, exhibit scalars α, β , not both zero, with $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$; then describe the span as a set, give a basis for it, and state its dimension.

	$\mathbf{u}$	$\mathbf{v}$		$\mathbf{u}$	$\mathbf{v}$
(a)	$[3, -6]$	$[-1, 2]$	(d)	$[\frac{1}{2}, \frac{3}{4}]$	$[2, 3]$
(b)	$[2, 5]$	$[5, 2]$	(e)	$[\sqrt{2}, 1]$	$[2, \sqrt{2}]$
(c)	$[0, 0]$	$[4, 1]$	(f)	$[3, 1]$	$[-1, 3]$

- (g) Find every k for which $[k, 2]$ and $[8, k]$ are dependent, and for each such k give the equation of the line they span.

Problem C.3 (A spanning set that is not a basis of $\mathbb{R}^2$).

Let $S = \{[1, 2], [3, 1], [4, 3]\}$, a set of three vectors in $\mathbb{R}^2$. It spans $\mathbb{R}^2$, but it is not a basis of $\mathbb{R}^2$.

- (a) Show that $[4, 3] = [1, 2] + [3, 1]$, and use it to write a combination $a[1, 2] + b[3, 1] + c[4, 3] = \mathbf{0}$ with a, b, c not all zero.
- (b) Write $\mathbf{w} = [7, 4]$ as a combination of the three vectors in two different ways.
- (c) Show that for *every* real number t ,

$$(1 - t)[1, 2] + (2 - t)[3, 1] + t[4, 3] = [7, 4].$$

So $\mathbf{w}$ has infinitely many sets of “coordinates” against S .

- (d) Which of the three two-element subsets of S are bases of $\mathbb{R}^2$?
- (e) In two sentences: S spans $\mathbb{R}^2$, so why is it not a basis? Which half of “basis” — existence of coordinates, or their uniqueness — does it fail?

Problem C.4 (Subspace or not?).

For each subset of $\mathbb{R}^2$, decide whether it is a subspace of $\mathbb{R}^2$. If it is, prove the three conditions, give a basis, and state its dimension. If it is not, name a condition that fails and give explicit vectors (and a scalar, if needed) that show it.

- (a) $\{(x, y) : 3x - 5y = 0\}$
- (b) $\{(x, y) : x + y = 1\}$
- (c) $\{(x, y) : xy = 0\}$
- (d) $\{(x, y) : y = x^2\}$
- (e) $\{(x, y) : x \text{ and } y \text{ are both integers}\}$
- (f) $\{(x, y) : 2x = y \text{ and } x = 3y\}$
- (g) $\{(x, y) : x^2 + y^2 \leq 1\}$

Problem C.5 (Three short proofs about span and independence).

Work from the definitions. Do not write vectors in components.

- (a) Prove: if $\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$, then $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u}, \mathbf{v}\}$. Prove both inclusions. (This is the general form of §18.6, where dropping a redundant vector cost nothing.)
- (b) Prove: if $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, then so are $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.
- (c) Prove the converse of (b): if $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are linearly independent, then so are $\mathbf{u}$ and $\mathbf{v}$.

Hint. Write $\mathbf{a} = \mathbf{u} + \mathbf{v}$ and $\mathbf{b} = \mathbf{u} - \mathbf{v}$, express $\mathbf{u}$ and $\mathbf{v}$ in terms of $\mathbf{a}$ and $\mathbf{b}$, and substitute into $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$.

Problem C.6 (Intersections and unions of subspaces).

Let S and T be subspaces of $\mathbb{R}^2$.

- (a) Prove that the intersection $S \cap T$ is a subspace.
- (b) Prove: if $S \cup T$ is a subspace, then $S \subseteq T$ or $T \subseteq S$.

Hint. Suppose neither holds, and pick $\mathbf{s} \in S$ with $\mathbf{s} \notin T$ and $\mathbf{t} \in T$ with $\mathbf{t} \notin S$. Then $\mathbf{s} + \mathbf{t}$ lies in $S \cup T$, so it lies in S or in T . Show that either case leads to a contradiction.

Problem C.7 (Spans in $\mathbb{R}^3$).

Let $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [0, 1, 1]$.

- (a) Show that $\mathbf{a}$ and $\mathbf{b}$ are linearly independent.
- (b) Decide whether each of $[1, 2, 1]$, $[1, 0, 0]$ and $[2, -1, -3]$ lies in $\text{span}\{\mathbf{a}, \mathbf{b}\}$. For each that does, give the coefficients. For each that does not, show that the system has no solution.

- (c) Show that $\text{span}\{\mathbf{a}, \mathbf{b}\} = \{[x, y, z] : y = x + z\}$. What kind of geometric object is it, and what is its dimension?
- (d) Let $\mathbf{c} = [1, 0, 0]$. Using the definition in the table of §24.2, show that $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are linearly independent, and write $[3, 5, 4]$ as a combination of them.
- (e) Is the plane $z = 1$ a subspace of $\mathbb{R}^3$? One sentence.

Part D. Orthogonality and the Pythagorean identity

Vector Unit, Part 5. Nonzero, mutually orthogonal vectors are linearly independent (§21.2). Against an orthogonal basis, coordinates need no system: the coefficient of each basis vector $\mathbf{b}$ is $\langle \mathbf{b}, \mathbf{w} \rangle / \|\mathbf{b}\|^2$, and $\mathbf{w}$ is the sum of its shadows on the basis vectors (§21.6). And if $\mathbf{u} \perp \mathbf{v}$ then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ (§23).

In $\mathbb{R}^3$, §21.4 proves that three mutually orthogonal nonzero vectors are independent, but only *announces* that they span. So wherever a problem below asks for coordinates in $\mathbb{R}^3$, substituting them back is part of the answer: that step is what shows the vector really is that combination.

Problem D.1 (Coordinates by projection alone).

- (a) Let $\mathbf{u} = [3, 4]$ and $\mathbf{v} = [-8, 6]$. Check that they are nonzero and orthogonal, and say which result makes them a basis of $\mathbb{R}^2$. Then find the coordinates of $[2, 11]$, $[-1, 7]$ and $[5, 0]$ with one dot product and one division per coordinate, and check each by substituting back.
- (b) For $[5, 0]$, find the coordinates again with the formula of §18.5. You should get the same numbers with noticeably more work.
- (c) Let $\mathbf{u} = [1, 1, 1]$, $\mathbf{v} = [1, -1, 0]$ and $\mathbf{w} = [1, 1, -2]$. Check that they are mutually orthogonal and nonzero. Find the coefficients of $\mathbf{x} = [3, 5, -2]$ and of $\mathbf{y} = [0, 0, 6]$ by projection, and substitute back.
- (d) Write $\mathbf{x}$ from (c) as the sum of its three shadows $\text{proj}_{\mathbf{u}}(\mathbf{x}) + \text{proj}_{\mathbf{v}}(\mathbf{x}) + \text{proj}_{\mathbf{w}}(\mathbf{x})$, listing each shadow as a vector.

Problem D.2 (Manufacturing an orthogonal basis).

The *lookahead* at the end of §21.6 promised that any basis can be converted into an orthogonal one by subtracting projections. Here is the first step of that procedure.

- (a) Let $\mathbf{u} = [3, 1]$ and $\mathbf{v} = [2, 4]$. Show that they are independent but not orthogonal. Compute $\mathbf{v}' = \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ and check that $\mathbf{v}' \perp \mathbf{u}$.
- (b) Find the coordinates of $\mathbf{w} = [7, 9]$ against the orthogonal basis $\{\mathbf{u}, \mathbf{v}'\}$ by projection. Then use $\mathbf{v}' = \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ to rewrite the result as a combination of the original $\mathbf{u}$ and $\mathbf{v}$, and check it.
- (c) Explain why $\text{span}\{\mathbf{u}, \mathbf{v}'\} = \text{span}\{\mathbf{u}, \mathbf{v}\}$: show that each of $\mathbf{v}'$ and $\mathbf{v}$ is a combination of the other pair.

Problem D.3 (Projection and the squared cosine).

Prove that for $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$,

$$\frac{\|\text{proj}_{\mathbf{u}}(\mathbf{v})\|^2}{\|\mathbf{v}\|^2} = \cos^2(\mathbf{u}, \mathbf{v}),$$

and evaluate both sides for $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [4, 7]$. Read the left side as “the fraction of $\mathbf{v}$ ’s squared length explained by the direction $\mathbf{u}$ ”. For this pair, is it close to 1 or to 0, and what does that say about the angle?

Problem D.4 (Pythagoras for three vectors, and for n).

Work from the laws alone (§23.2 does), so that the result holds in any $\mathbb{R}^n$.

- (a) Prove: if $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are mutually orthogonal, then $\|\mathbf{u} + \mathbf{v} + \mathbf{w}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$.
- (b) Deduce: if $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are orthonormal, then $\|\alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}\|^2 = \alpha^2 + \beta^2 + \gamma^2$. In words: against an orthonormal basis, the squared length of a vector is the sum of the squares of its coordinates.
- (c) State, in Σ notation and without proof, the version of (a) for k mutually orthogonal vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$.

Problem D.5 (Everything orthogonal to a vector).

- (a) Prove: if $\mathbf{w} \perp \mathbf{u}$ and $\mathbf{w} \perp \mathbf{v}$, then $\mathbf{w}$ is orthogonal to every vector in $\text{span}\{\mathbf{u}, \mathbf{v}\}$.
- (b) For a vector $\mathbf{u} \in \mathbb{R}^2$, let $\mathbf{u}^\perp = \{\mathbf{x} \in \mathbb{R}^2 : \langle \mathbf{u}, \mathbf{x} \rangle = 0\}$. Prove that $\mathbf{u}^\perp$ is a subspace of $\mathbb{R}^2$.

Part E. Computing in $\mathbb{R}^n$

Vector Unit, §20.3 and §24. Every definition of the unit holds in $\mathbb{R}^n$ with a Σ in it (the table in §24.2). The ones used most below:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i, \quad \|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}, \quad \|\mathbf{v}\|_\infty = \max_{1 \leq i \leq n} |v_i|, \quad \cos(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Write $\mathbf{e}_i$ for the standard basis vector with a 1 in slot i , and $\mathbf{1} = [1, 1, \dots, 1]$ for the all-ones vector in $\mathbb{R}^n$. In $\mathbb{R}^n$ the angle between nonzero vectors is *defined* as $\theta = \arccos \cos(\mathbf{u}, \mathbf{v}) \in [0, \pi]$ (§24.5). The sum formulas $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ and $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ are in the foundations review.

Cauchy–Schwarz holds in $\mathbb{R}^n$ (§24.3). Your proof of Homework 1 Problem E.2 used only **D1–D4**, so it already proves it.

Problem E.1 (Arithmetic in $\mathbb{R}^4$).

Let

$$\mathbf{u} = [1, -2, 0, 3], \quad \mathbf{v} = [2, 1, -1, 0], \quad \mathbf{w} = [0, 3, 2, -1], \quad \alpha = -2.$$

- (a) $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $\alpha\mathbf{w}$, and $2\mathbf{u} - \mathbf{v} + 3\mathbf{w}$.
- (b) $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, $\|\mathbf{w}\|$, exactly.
- (c) $\|\mathbf{u}\|_1$, $\|\mathbf{u}\|_\infty$, $\|\mathbf{w}\|_1$, $\|\mathbf{w}\|_\infty$.

- (d) $\langle \mathbf{u}, \mathbf{v} \rangle$, $\langle \mathbf{u}, \mathbf{w} \rangle$, $\langle \mathbf{v}, \mathbf{w} \rangle$. Which pair is orthogonal?
- (e) $\cos(\mathbf{u}, \mathbf{w})$ and $\cos(\mathbf{v}, \mathbf{w})$, exactly, and the angle between $\mathbf{u}$ and $\mathbf{w}$ in degrees.
- (f) $d_2(\mathbf{u}, \mathbf{w})$, $d_1(\mathbf{u}, \mathbf{w})$ and $d_\infty(\mathbf{u}, \mathbf{w})$, and confirm they come out in the order §14.8 predicts.
- (g) $\mathbf{u}$ written as a combination of $\mathbf{e}_1, \dots, \mathbf{e}_4$, and $\langle \mathbf{e}_4, \mathbf{u} \rangle$.

Problem E.2 (The dot-product laws in $\mathbb{R}^n$).

§24.3 says the laws survive in $\mathbb{R}^n$ because the proofs “run over n slots instead of 2”. Run them. Use Σ notation throughout, and at each step name the fact you use: a property of real numbers, or one of the rules for sums in the foundations review (splitting a sum, pulling out a constant).

- (a) **D1:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
- (b) **D2:** $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.
- (c) **D3:** $\langle \mathbf{u}, \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle$.
- (d) **D4:** $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, with equality only for $\mathbf{u} = \mathbf{0}$.
- (e) **N2** and **N3** for the 1-norm on $\mathbb{R}^n$: $\|\alpha \mathbf{v}\|_1 = |\alpha| \|\mathbf{v}\|_1$ and $\|\mathbf{u} + \mathbf{v}\|_1 \leq \|\mathbf{u}\|_1 + \|\mathbf{v}\|_1$.

Problem E.3 (Norms of patterned vectors).

Let $n \geq 1$. Give each answer as a formula in n .

- (a) $\|\mathbf{1}\|_1$, $\|\mathbf{1}\|_2$, $\|\mathbf{1}\|_\infty$.
- (b) Let $\mathbf{v} = [1, 2, 3, \dots, n]$. Compute $\|\mathbf{v}\|_1$, $\|\mathbf{v}\|_2$, $\|\mathbf{v}\|_\infty$ and $\langle \mathbf{1}, \mathbf{v} \rangle$.
- (c) With $\mathbf{v}$ as in (b), show that

$$\cos(\mathbf{1}, \mathbf{v}) = \frac{\sqrt{6}}{2} \sqrt{\frac{n+1}{2n+1}}.$$

Evaluate the angle between $\mathbf{1}$ and $\mathbf{v}$ in degrees for $n = 1, 2$ and 10 , and find its limit as $n \rightarrow \infty$.

- (d) For $i \neq j$, compute $\|\mathbf{e}_i - \mathbf{e}_j\|_1$, $\|\mathbf{e}_i - \mathbf{e}_j\|_2$ and $\|\mathbf{e}_i - \mathbf{e}_j\|_\infty$. Why does the answer not depend on n, i or j ?

Problem E.4 (Cosine similarity for documents).

A common way to compare documents is to count words. Fix a vocabulary of five words — *vector*, *matrix*, *norm*, *pizza*, *soccer* — and record each document as a vector in $\mathbb{R}^5$ of word counts:

Document	Counts
<i>A</i>	[3, 1, 4, 0, 0]
<i>B</i>	[6, 2, 8, 0, 0]
<i>C</i>	[0, 0, 0, 1, 5]
<i>D</i>	[2, 0, 3, 1, 0]

- (a) Compute $\cos(A, B)$, $\cos(A, C)$, $\cos(A, D)$ and $\cos(C, D)$ exactly, and the last two to four decimal places.
- (b) Compute the Euclidean distances $d_2(A, B)$, $d_2(A, C)$, $d_2(A, D)$.
- (c) Which document is most similar to A by cosine similarity? Which is nearest to A by Euclidean distance? Document B is document A pasted in twice. Explain which of the two measures gives the sensible answer for comparing topics, and why.

Problem E.5 (The mean and the variance are a projection).

A data set of n numbers is a vector $\mathbf{x} \in \mathbb{R}^n$. Take $\mathbf{x} = [4, 7, 1, 8, 5]$.

- (a) Compute $\text{proj}_{\mathbf{1}}(\mathbf{x})$. Then prove, for any $\mathbf{x} \in \mathbb{R}^n$, that $\text{proj}_{\mathbf{1}}(\mathbf{x}) = \bar{x} \mathbf{1}$, where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is the mean. Projecting onto $\mathbf{1}$ replaces every entry by the average.
- (b) The orthogonal piece $\mathbf{x} - \bar{x} \mathbf{1}$ is the vector of *deviations from the mean*. Compute it, check that it is orthogonal to $\mathbf{1}$, and deduce the general fact that deviations from the mean sum to zero.
- (c) The (population) **variance** is $\sigma^2 = \frac{1}{n} \|\mathbf{x} - \bar{x} \mathbf{1}\|^2$ and the **standard deviation** is σ . Compute both.
- (d) For a second data set $\mathbf{y}$, the **correlation coefficient** of $\mathbf{x}$ and $\mathbf{y}$ is the cosine similarity of their deviation vectors, $r = \cos(\mathbf{x} - \bar{x} \mathbf{1}, \mathbf{y} - \bar{y} \mathbf{1})$. Compute r exactly and to four decimal places for $\mathbf{y} = [3, 5, 1, 6, 5]$.

Problem E.6 (The standard basis, and the diagonals of a cube).

- (a) Show that $\langle \mathbf{e}_i, \mathbf{v} \rangle = v_i$ for every $\mathbf{v} \in \mathbb{R}^n$, and that $\langle \mathbf{e}_i, \mathbf{e}_j \rangle$ is 1 when $i = j$ and 0 otherwise.
- (b) Show that the cosine of the angle between the diagonal $\mathbf{1}$ and the edge direction $\mathbf{e}_1$ is $1/\sqrt{n}$. Give the angle in degrees for $n = 2, 3, 4$ and 100, and find its limit as $n \rightarrow \infty$. Say in one sentence what this means for the long diagonal of a high-dimensional cube.

Problem E.7 (Comparing the norms in $\mathbb{R}^n$).

Prove each inequality for every $\mathbf{v} \in \mathbb{R}^n$. Work from the Σ formulas.

- (a) $\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1$.

Hint. For the second, compare $\|\mathbf{v}\|_1^2 = (\sum |v_i|)^2$ with $\sum v_i^2$. What do the cross terms contribute?

- (b) $\|\mathbf{v}\|_1 \leq n \|\mathbf{v}\|_\infty$ and $\|\mathbf{v}\|_2 \leq \sqrt{n} \|\mathbf{v}\|_\infty$.

- (c) $\|\mathbf{v}\|_1 \leq \sqrt{n} \|\mathbf{v}\|_2$.

Hint. Apply Cauchy–Schwarz in $\mathbb{R}^n$ to the vectors $[|v_1|, \dots, |v_n|]$ and $\mathbf{1}$.

In one sentence: together these say that in $\mathbb{R}^n$ any two of the three norms are within a factor of n of each other. What changes as n grows?

Part F. Derivatives, partial derivatives, and the gradient

Vector Calculus Unit, Part 1. In this part, a bare section number such as §6.1 refers to the *vector calculus* unit, whose numbering restarts at 1. References into the vector unit say so. Throughout,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad \frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right],$$

and the exponential and logarithm laws are **E1–E5** and **L1–L5** of §3.

Problem F.1 (The rules, in one variable).

Differentiate each function, and name the rule or rules you use — those of §2, and the derivatives of e^x and $\ln x$ found in §4. Simplify where it helps.

- (a) $x^3 e^x$
- (b) $\ln(x^2 + 1)$
- (c) $e^{-x^2/2}$
- (d) $\frac{\sin x}{1 + e^x}$
- (e) $\sqrt{x} \ln x$
- (f) $\ln(\ln x)$
- (g) $e^{3x} \tan x$
- (h) $\sec(x^2)$

Problem F.2 (Exponential and logarithm algebra).

- (a) Simplify each expression, naming the law of §3 used at each step:

$$\ln(e^3 e^{-5}), \quad e^{2 \ln 3}, \quad \ln \sqrt{e}, \quad \ln 8 - 3 \ln 2, \quad e^{\ln 5 - \ln 2}.$$

- (b) Solve $e^{2x} - 3e^x + 2 = 0$.

Hint. It is a quadratic in the unknown e^x .

- (c) Solve $\ln x + \ln(x-3) = \ln 4$. One solution of the quadratic you reach must be rejected. Which one, and why?

Problem F.3 (Two ways, again).

§4.2 and §4.3 found the derivative of e^x two ways: by implicit differentiation, and from the definition. Do the same kind of thing here.

- (a) Let $a > 0$. Derive $\frac{d}{dx} a^x = a^x \ln a$ in two ways:

- (i) write $a^x = e^{x \ln a}$, justifying it by the inverse relation $e^{\ln a} = a$ and **E4**, then use the chain rule;

- (ii) put $y = a^x$, take $\ln$ of both sides using **L4**, and differentiate implicitly.
- (b) §4.2 went from $\ln$ to $\exp$. Go the other way: taking $\frac{d}{dx}e^x = e^x$ as known, derive $\frac{d}{dx}\ln x = 1/x$ by writing $y = \ln x$ as $e^y = x$ and differentiating implicitly.

Problem F.4 (Partial derivatives).

Compute $\partial f/\partial x$ and $\partial f/\partial y$. Assume each function is considered where it is defined.

(a) $f(x, y) = 3x^2y^3 - 4xy + 7$

(b) $f(x, y) = x e^{xy}$

(c) $f(x, y) = \ln(x^2 + y^2)$

(d) $f(x, y) = \sin x \cos y$

(e) $f(x, y) = x/y$

(f) $f(x, y) = (x - 2y)^5$

(g) $f(x, y) = e^x \ln y$

(h) $f(x, y) = \sqrt{x^2 + y^2}$

Watch out. In (b), the variable held constant appears in two places — in front, and inside the exponent. Holding y fixed, f is a product of two functions of x , so the product rule is needed. Holding x fixed, the front x is a constant.

Problem F.5 (Partial derivatives in context).

- (a) The volume of a cylinder of radius r and height h is $V(r, h) = \pi r^2 h$. Compute $\partial V/\partial r$ and $\partial V/\partial h$, evaluate both at $r = 3$ cm, $h = 10$ cm, and give their units. Use them to estimate which adds more volume: widening the radius by 1 mm or raising the height by 1 mm. Then compute the two actual changes and compare.
- (b) For a fixed amount of gas, pressure depends on temperature T (in kelvin) and volume V (in litres) as $P(T, V) = 8T/V$ (in kilopascals). Compute P , $\partial P/\partial T$ and $\partial P/\partial V$ at $T = 300$, $V = 10$, with units. Explain the sign of each partial derivative physically.

Problem F.6 (The gradient).

- (a) Let $f(x, y) = x^3 - 3xy + y^2$. Compute ∇f , evaluate it at $(1, 2)$, $(2, -1)$ and $(0, 0)$, and compute $\|\nabla f(2, -1)\|$.
- (b) Find every point where $\nabla f = \mathbf{0}$ for the f of (a).
- (c) Let $f(x, y) = \ln(x^2 + y^2)$. Show that $\|\nabla f(x, y)\| = 2/\|[x, y]\|$, and describe the direction of ∇f .
- (d) Let $f(x, y) = \sqrt{x^2 + y^2} = \|[x, y]\|$. Show that $\nabla f(x, y)$ is the *normalization* of $[x, y]$ (vector unit §12.1), so that $\|\nabla f\| = 1$ at every point other than the origin. Why is ∇f undefined at the origin?

Questions about any problem here belong in the class Discord, where everyone can see the answer. Questions that are specific to your own work — “is my proof of C.6(b) valid?” — are better brought to office hours, Thursdays 3:00–4:00 PM in Kiely Hall 508. Please email ahead.